Geometry
Unit 13
Lesson 9 Homework

Name__Answer Key_______
Date___________________
Period__________________

1. Samal is using a computer software to design the layout of his new game room. The base of the new game room creates $GAMR$ with coordinates $G(-7,-5)$, $A(3,-9)$, $M(5,-4)$, and $R(-5,0)$.

a. Complete the table. Show your work in the table. Leave answers in exact form.

Line Segment	Length
$\overline{GA}$	$\sqrt{(3-(-7))^2+(-9-(-5))^2}=\sqrt{10^2+(-4)^2}=\sqrt{100+16}$ $=\sqrt{116}=2\sqrt{29}$
$\overline{AM}$	$\sqrt{(-9-(-4))^2+(3-5)^2}=\sqrt{(-5)^2+(-2)^2}=\sqrt{25+4}=\sqrt{29}$
$\overline{MR}$	$\sqrt{(0-(-4))^2+(-5-5)^2}=\sqrt{4^2+(-10)^2}=\sqrt{16+100}=\sqrt{116}$ $=2\sqrt{29}$
$\overline{GR}$	$\sqrt{(0-(-5))^2+(-5-(-7))^2}=\sqrt{5^2+2^2}=\sqrt{25+4}=\sqrt{29}$

b. Complete the table. Show your work in the table. Leave answers in exact form.

Diagonal Line Segment	Length
$\overline{GM}$	$\sqrt{(-5-(-4))^2+(-7-5)^2}=\sqrt{(-1)^2+(-12)^2}=\sqrt{1+144}$ $=\sqrt{145}$
$\overline{AR}$	$\sqrt{(-9-0)^2+(3-(-5))^2}=\sqrt{(-9)^2+8^2}=\sqrt{81+64}=\sqrt{145}$

c. Finish the statement.

$GAMR$ is classified as a rectangle because … opposite sides are congruent and diagonals are congruent.

d. If 1 unit on the coordinate plane equals 4 feet (ft), what are the dimensions of the base of the new game room? Round to the nearest thousandth.
$GA=2\sqrt{29}\approx(10.7703)4=43.081$ ft
$AM=\sqrt{29}\approx(5.38516)4=21.541$ ft

2. $STLE$ has vertices at $S(-5,6)$, $T(0,-3)$, $L(6,-5)$, and $E(1,4)$.

$STLE$ is classified as a(n) parallelogram because... opposite sides congruent, but diagonals are not.

3. Franco is designing a template for his new goat pen. He created two quadrilaterals using a computer software program.

$MFCR$	$AMBT$
$M(-1,5)$, $F(-9,-3)$, $C(-5,-7)$, $R(3,1)$	$A(-2,-3)$, $M(4,5)$, $B(0,7)$, $T(-6,0)$

a. Which quadrilateral can be classified as a rectangle? Show your work.
$MFCR$
$MR = \sqrt{(3-(-1))^2 + (1-5)^2} = \sqrt{4^2+4^2} = \sqrt{16+16} = 4\sqrt{2}$, slope $= -1$
$FC = \sqrt{(-7-(-3))^2 + (-5-(-9))^2} = \sqrt{4^2+4^2} = \sqrt{16+16} = 4\sqrt{2}$, slope $= -1$
$CR = \sqrt{(-7-1)^2 + (-5-3)^2} = \sqrt{8^2+8^2} = \sqrt{64+64} = 8\sqrt{2}$, slope $= 1$

b. List the properties of a rectangle that you used.
Opposite sides congruent and parallel, $\overline{MR} \parallel \overline{FC}$, $\overline{MR} \cong \overline{FC}$, adjacent sides perpendicular by comparing slope.

c. If 1 unit of the coordinate plane equals 7 feet (ft), what are the measurements of the sides of the rectangle?
$7(4\sqrt{2}) = 28\sqrt{2} \approx 39.6$ ft
$7(8\sqrt{2}) = 56\sqrt{2} \approx 79.2$ ft

4. Akilah was graphing square $STVE$, but mixed up her vertices. She knows that point S has coordinates $(0,4)$ and point E has coordinates $(8,0)$. Her two options for points T and V are shown.

Point T	Point V
$(-4,-4)$	$(-2,-8)$
$(4,-4)$	$(4,-8)$

a. Which coordinates are correct for points T and V? Justify your answer or show your work.
$SE = 4\sqrt{5}$, slope $= -\frac{1}{2}$
$ST = 4\sqrt{5}$, slope $= 2$
$EV = 4\sqrt{5}$, slope $= 2$
$T(-4,-4)$, $V(4,-8)$

b. Which properties of a square did you use to determine the correct coordinates for points T and V?
Length of all 4 sides are congruent and slope of adjacent sides opposite reciprocal (sides perpendicular).